

Binisha Chapagain

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Summary

Django-focused backend developer with hands-on experience building scalable web applications and intelligent systems. Skilled in developing RESTful APIs, authentication systems, and role-based platforms using Django, along with integrating AI features such as LLM-powered applications, Retrieval-Augmented Generation (RAG), and NLP pipelines using LangChain and LangGraph.

Experienced in designing data-driven backend services, working with PostgreSQL, and managing asynchronous tasks using Celery and Redis. Passionate about building efficient, real-world web applications with clean architecture and scalable design.

Seeking a Django Internship to apply backend development skills while integrating AI-driven features into modern web applications.

Skills

Programming Languages: Python, SQL, Java, PHP, C

Machine Learning and AI: Machine Learning, Deep Learning, Neural Network, Transformers, Natural Language Processing(NLP), Generative AI, Agentic AI, Retrieval-Augmented Generation(RAG)

Frameworks and Libraries: NumPy, Pandas, scikit-learn, TensorFlow, Keras, NLTK, HuggingFace, LangChain, LangGraph, Django, Bootstrap, Tailwind CSS

Backend and APIs: FastAPI, REST APIs, SerpAPI

Databases and Vector Stores: MySQL, PostgreSQL, SQLite, ChromaDB, Elasticsearch

Tools and Platforms: Git, GitHub, Docker, Jupyter Notebook, VS Code, Google Colab, Redis, Celery

Web and Development: Streamlit, Tailwind CSS

Experience

AI/ML Intern

Nov 2025 – Feb 2026

Inspiring Lab Pvt. Ltd.

- Developed AI-powered systems using Agentic AI architectures and Retrieval-Augmented Generation (RAG).
- Built scalable backend APIs using FastAPI and PostgreSQL for AI-driven applications.
- Implemented multi-agent orchestration workflows using LangGraph and Google ADK.
- Designed and implemented an AI-powered blog generation pipeline integrating SerpAPI for real-time search and SEO-based topic discovery.
- Built vector search systems using Elasticsearch for semantic retrieval and content indexing.
- Integrated Redis for conversational memory and Celery for asynchronous background task processing.
- Developed integrations with Shopify and WooCommerce APIs to automate e-commerce workflows.
- Built an AI-driven sales support chatbot capable of context-aware customer interaction.

Projects

• Rural Health AI – Smart Healthcare Solutions

Developed a full-stack healthcare platform using Django as the core backend framework, enabling scalable APIs, authentication, and role-based access for patients, health workers, and administrators. Designed an Agentic RAG system to intelligently process clinical queries, implementing a custom 3-tier intent classification (Heuristic, LLM, Hybrid) for efficient routing and decision-making.

Built multi-step reasoning agents with LangGraph and adaptive retrieval pipelines, integrating ChromaDB semantic search with a SerpAPI-based web fallback to improve response reliability under low-confidence scenarios. Engineered an AI-driven medical scheduling system with urgency triage using a Min-Heap Priority Queue, achieving efficient $O(\log n)$ prioritization of critical cases.

Integrated Celery and Redis within the Django ecosystem to handle asynchronous background tasks such as automated appointment scheduling, reminders, and no-show cancellations.

Tools: Django, LangGraph, ChromaDB, Gemini, SerpAPI, Celery, Redis, PostgreSQL

- **Text-to-SQL Agent Using LangGraph**

Designed an intelligent agent to convert natural language questions into SQL queries and return human-readable answers. Built a secure, multi-step workflow for table listing, schema retrieval, query generation, and result interpretation. Integrated a user-friendly Streamlit interface with logging and error handling for smooth interactions.

Tools: Python, LangChain, LangGraph, Groq LLM, Streamlit, SQLite

GitHub: github.com/BinishaChapagain/text-to-sql-agent

- **AI-Powered Document Q&A Agent**

Implemented a Retrieval-Augmented Generation (RAG) system for document question-answering with PDF, TXT, and DOCX support and web search fallback. Built a multi-agent workflow using LangGraph, with a React frontend and FastAPI backend, leveraging Groq LLaMA and ChromaDB for fast LLM inference and persistent vector storage.

Tools: Python, FastAPI, React, LangGraph, LangChain, Groq, ChromaDB, SerpAPI

GitHub: github.com/BinishaChapagain/AIPoweredDocumentQ&AAgent

- **Intelligent Multi-Document Chatbot Using LlamaParse**

Designed a robust chatbot system that processes and understands multi-format documents like pdf, docx, txt for intelligent question-answering. Implemented LlamaParse for accurate document parsing, HuggingFace embeddings for semantic text representation, and ChromaDB for efficient vector storage. Integrated Groq's LLaMA-3.1 model with LangChain to deliver fast, context-aware responses while preserving chat history for seamless conversations.

Tools: Python, LlamaParse, HuggingFace, ChromaDB, LangChain, Groq

GitHub: github.com/BinishaChapagain/multidocumentchatbot

- **RNN-Based Sentiment Analysis**

Implemented a sentiment analysis system to classify IMDB movie reviews as positive or negative using a Simple RNN architecture. Designed and trained a neural network with embedding and recurrent layers to capture temporal patterns in text data, achieving approximately 85% accuracy with binary cross-entropy loss and the Adam optimizer. Integrated an interactive Streamlit interface that enables real-time sentiment prediction from user input. Applied early stopping during training to improve model generalization and reduce overfitting.

Tools: Python, TensorFlow, Keras, NumPy, Streamlit

GitHub: github.com/BinishaChapagain/RNNBasedSentimentAnalysis

- **House Price Prediction Using Linear Regression**

Built a robust supervised machine learning model to accurately predict house prices based on key features such as location, property size, and number of bedrooms. Designed and implemented a complete ML pipeline including data cleaning, feature engineering, model training, and performance evaluation.

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

GitHub: github.com/BinishaChapagain/housepriceprediction

Education

Lumbini ICT Campus, Tribhuvan University, Nepal

Bachelor of Computer Application

Jan 2022 – Present

Courses: Operating System, Data Structures, Object Oriented Programming, Database, Machine Learning, Artificial Intelligence.

Training

AI and Machine Learning Practical Training

2025

Mentor by Abinash Pant

Completed hands-on training on building intelligent systems using Machine Learning and Generative AI. Gained practical experience with Retrieval-Augmented Generation, LangChain, vector databases, and autonomous AI agents. Developed projects including a PDF-interactive AI chatbot, a Multi-Agent conversational system, and a chatbot using LlamaParse, applying concepts in Machine Learning, Deep Learning, and AI applications.

Machine Learning and AI Online Courses from Udemy

2025

Completed comprehensive online courses covering supervised and unsupervised learning, deep learning, and AI-powered applications. Applied theoretical knowledge in practical projects using Python, TensorFlow, and relevant machine learning frameworks, strengthening hands-on skills in developing AI and ML solutions.